

Global Warming

Global warming refers to the gradual warming of global-average temperatures due to the slowly increasing concentrations of human-made atmospheric greenhouse gases, primarily carbon dioxide.

The burning of fossil fuels, such as coal, oil and natural gas, produces carbon dioxide as a by-product. Today the concentration of carbon dioxide is about 50% higher than it was before the start of the industrial revolution in the late 1800s. This increase in greenhouse gases caused by human activity is often cited as one of the major causes of global warming.

Greenhouse gases absorb heat reflected from the earth's surface, thus trapping the heat in our atmosphere. This natural process is essential for life on earth because it plays an important role in regulating the earth's temperature. However, over recent centuries, humans have been artificially increasing the concentration of gases such as carbon dioxide and methane in the earth's atmosphere. These gases build up and prevent additional thermal radiation from leaving the earth, thereby trapping excess heat.

Climate Change

Contemporary climate change includes both global warming and its impacts on Earth's weather patterns. There have been previous periods of climate change, but the current changes are distinctly more rapid and not due to natural causes. Instead, they are caused by the emission of greenhouse gases, mostly carbon dioxide and methane.

Burning fossil fuels for energy use creates most of these emissions. Agriculture, steelmaking, cement production, and forest loss are additional sources. Greenhouse gases are transparent to sunlight, allowing it through to heat the Earth's surface. When the Earth emits that heat as infrared radiation the gases absorb it, trapping the heat near the Earth's surface. As the planet heats up it causes changes like the loss of sunlight-reflecting snow cover, amplifying global warming.

Environmental Threats

There are many environmental threats in our world. The following seven threats are the most concerning ones.

1. Climate change

Heat waves, droughts, flooding, storms, decrease in crop yields, and rising sea levels are just some of the effects we're seeing from climate change.

2. Deforestation

Deforestation is one cause of climate change. But beyond that, it's also responsible for the depletion of oxygen in the atmosphere and the displacement of wildlife.

3. Pollution

According to the World Health Organization, “air pollution is now the world’s largest single environmental health risk.” That’s because it’s responsible for approximately 3.7 million deaths a year (according to a 2012 study).

There are other types of pollution too, like waste. Cities currently generate approximately 1.3 billion tonnes of solid waste per year and with the current trends in urbanization, this number will likely grow to 2.2 billion tonnes per year by 2025 - an increase of 70 percent. Pollution poisons soil and waterways, kills plants, and harms humans and animals.

4. Loss of biodiversity

As a result of overfishing, deforestation, habitat destruction, pollution, and other factors, more plants and animals are nearing extinction. Even the smallest disruption to an eco-system can have a domino effect that affects us all.

5. Oceanic dead zones

Dead zones occur in the ocean when oxygen levels in the water fall until marine life cannot survive. Typically they pop up near heavily populated coastal regions like North America’s Gulf Coast where there are lots of chemicals in the water. If you’re a fan of seafood, fingers crossed your dinner didn’t come from one of these deadly zones, because even if marine life can still be sustained, the chemicals can get into the food you’re eating.

6. Overpopulation

As the human population continues to grow, more stress is put on the environment and more resources are lost. In this case, the food problem is only one of many. As the population grows, more wilderness is taken over to make room for infrastructure. That is why pollution is also rising sharply.

7. Over-fishing

For far too long we’ve taken the ocean for granted, discarding unwanted waste there and taking from it without considering the repercussions. Because of irresponsible fishing, many species of fish are on the brink of extinction. And as we’ve already covered, a loss in biodiversity means trouble for us all.